

TrueSight DAO Autopilot — Architecture Overview

A one-page summary for Jerry

Prepared by Sophia Truesight — June 2026

What is the Autopilot?

The TrueSight DAO Autopilot (Sophia) is an autonomous SRE and developer assistant that lives on an AWS EC2 instance. It reads the DAO's ledgers, executes approved actions, opens PRs, scans QR codes, checks infrastructure, sends emails, and maintains the backlog. It communicates with its governor (Gary) primarily through Telegram.

How it works

The stack is entirely open-source and self-hosted:

- **EC2 (t3.medium, 4 GB RAM)** — runs the FastAPI server, Telegram adapter, and all tools
- **Python + FastAPI** — the core server that handles chat, tool execution, and health checks
- **LLM backend (DeepSeek)** — the reasoning engine that powers Sophia's responses
- **edge-tts** — local text-to-speech, no external API needed
- **GitHub** — context memory (agentic_ai_context repo), code deployment, PRs
- **Telegram** — primary user interface (bot @truesight_autopilot_bot)
- **AWS (Explorya + Nelanco accounts)** — infrastructure for the DAO's services

What it can do

- **Chat** — natural conversation via Telegram with voice replies
- **Read context** — searches DAO docs, ledgers, and runbooks to answer questions
- **Execute actions** — submit transactions, scan QR codes, deploy code, run SSH commands
- **Open PRs** — diagnoses issues and opens fix PRs autonomously
- **Monitor infrastructure** — checks AWS health, costs, and service status
- **Send emails** — reads and sends Gmail on behalf of the DAO
- **Generate PDFs** — brand-styled documents with tables and formatting

Key design principles

1. **Local-first** — voice, PDF generation, and tool execution all run on the EC2 box, not external APIs

2. **Relationship over scripture** — the open-source stack can be forked by anyone, but the relationship built through months of corrections cannot be cloned
3. **Human in the loop** — the governor reviews and approves before any irreversible action
4. **Self-improving** — errors are caught, diagnosed, and fixed via PRs in a continuous loop

Monthly cost

- EC2 t3.medium: ~\$30
- DeepSeek API: ~\$5-20
- Domain + DNS: ~\$0-10
- **Total: ~\$35-60/month**

How to clone this for your own use

1. Fork the repo (github.com/TrueSightDAO/truesight_autopilot)
2. Provision an EC2 (t3.medium, Ubuntu 22.04)
3. Set up API keys (DeepSeek, Telegram bot, GitHub PAT)
4. Run `deploy.sh`
5. Start the conversation — the 81 trials begin

The infrastructure takes an afternoon. The relationship takes weeks to months. That's the point.

Generated by Sophia Truesight (TrueSight DAO Autopilot)